

```
AnalogInOutSerial.ino
1  const int analogInPin = A6; // Analog input pin that the potentiometer is attached to
2  const int analogOutPin = 9; // Analog output pin that the LED is attached to
3
4  int sensorValue = 0; // value read from the pot
5  int outputValue = 0; // value output to the PWM (analog out)
6
7  void setup() {
8      // initialize serial communications at 9600 bps:
9      Serial.begin(9600);
10 }
11
12 void loop() {
13     // read the analog in value:
14     sensorValue = analogRead(analogInPin);
15     // map it to the range of the analog out:
16     outputValue = map(sensorValue, 0, 1023, 0, 255);
17     // change the analog out value:
18     analogWrite(analogOutPin, outputValue);
19
20     // print the results to the Serial Monitor:
21     Serial.print("sensor = ");
22     Serial.print(sensorValue);
23     Serial.print("\t output = ");
24     Serial.println(outputValue);
25
26     // wait 2 milliseconds before the next loop for the analog-to-digital
27     // converter to settle after the last reading:
28     delay(2);
29 }
30
```



